

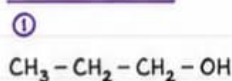
1. Alcohols (R-OH group)

→ Organic compounds in which -OH group is bonded to sp^3 hybridised carbon.

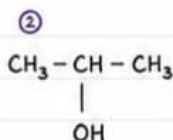
→ General formula : $R-OH$

Classification (on the basis of -OH group)

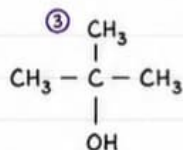
Primary (1°)	Secondary (2°)	Tertiary (3°)
-OH on 1° C $R-CH_2-OH$	-OH on 2° C $R-\underset{\substack{ \\ OH}}{CH}-R'$	-OH on 3° C $R-\underset{\substack{ \\ R''}}{C}-R'$ OH

Examples :

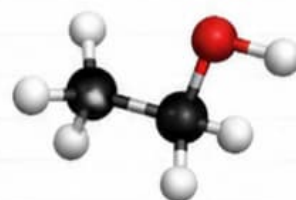
Propan-1-ol
(1° alcohol)



Propan-2-ol
(2° alcohol)



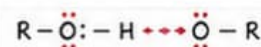
2-Methylpropan-2-ol
(3° alcohol)

3D Structure of Ethanol

CH_3-CH_2-OH

Physical Properties :

- Lower alcohols (C_1-C_3) are colourless, miscible with water.
- Solubility decreases with increase in molecular mass.
- Boiling point of alcohols > corresponding alkanes (due to H-bonding).
- They have characteristic pleasant odour.

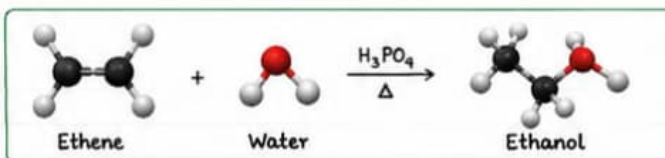
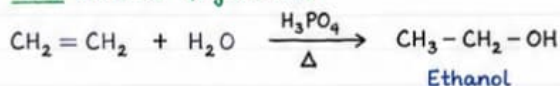
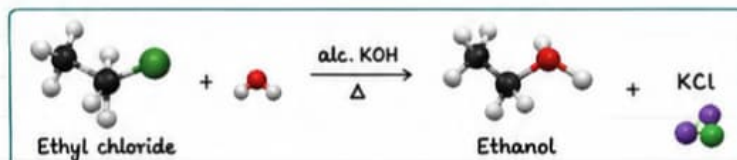
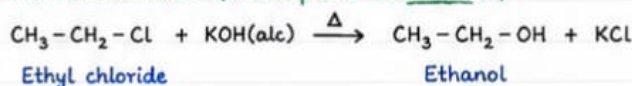
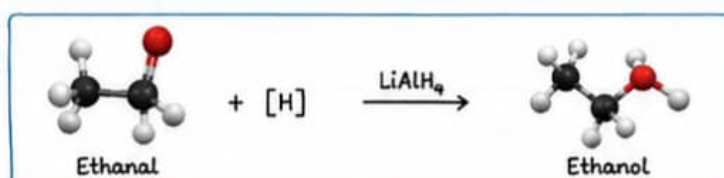
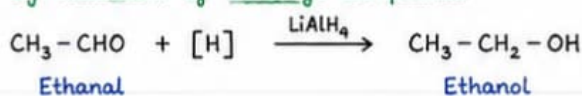
Hydrogen Bonding in Alcohols

(Stronger intermolecular force)

Nomenclature :

- IUPAC name : Replace 'e' of alkane with '-ol'.
- -OH group gets the lowest locant.
- Common name : Alkyl group name + alcohol.

Formula	IUPAC Name	Common Name
CH_3-OH	Methanol	Methyl alcohol
CH_3-CH_2-OH	Ethanol	Ethyl alcohol
$CH_3-CH_2-CH_2-OH$	Propan-1-ol	n-Propyl alcohol

Preparation of Alcohols :① **From Alkenes (Hydration)**② **From Haloalkanes (Nucleophilic substitution)**③ **By Reduction of Carbonyl Compounds****Key Points**

- ★ -OH group makes alcohols polar.
- ★ Lower alcohols are soluble in water.
- ★ Boiling point increases with molecular mass.
- ★ H-bonding is responsible for high boiling point.

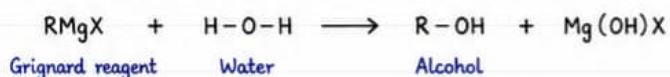
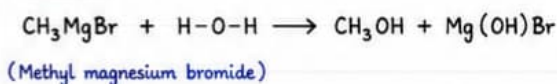
Remember

★ $1^\circ > 2^\circ > 3^\circ$
(Reactivity with HX)

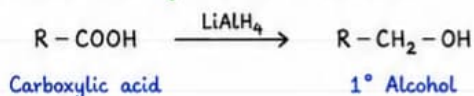


Preparation of Alcohols (Contd.):

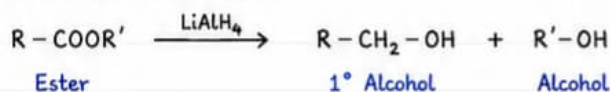
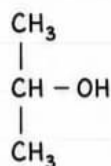
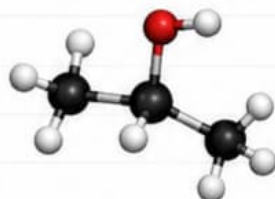
④ From Grignard Reagents

Example:

⑤ From Carboxylic Acids (Reduction)



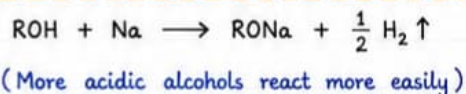
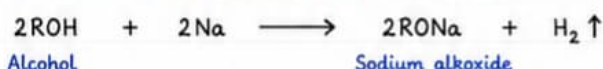
⑥ From Esters (Reduction)

3D Structure of Propan-2-ol

(Propan-2-ol / Isopropyl alcohol)

Chemical Properties of Alcohols:

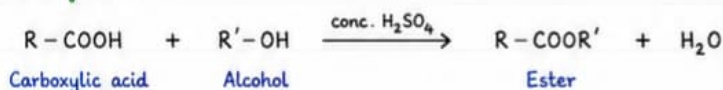
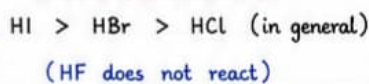
① With Active Metals (Formation of Alkoxides):



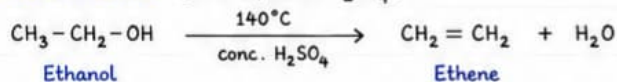
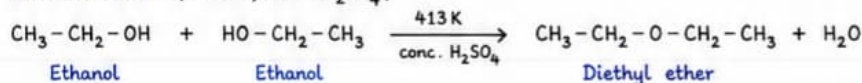
② With Hydrogen Halides:

Order of Reactivity with HX

③ Esterification:

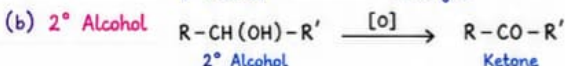
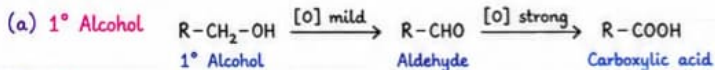
Reactivity towards HX

④ Dehydration (Elimination):

(a) Intramolecular (140°C, conc. H₂SO₄)(b) Intermolecular (413 K, conc. H₂SO₄)Important!

Dehydration of 1° alcohol at 140°C gives alkene, but at 413 K gives ether.

⑤ Oxidation:

Common Oxidising Agents

- PCC (Pyridinium chlorochromate)
- Acidified K₂Cr₂O₇
- KMnO₄ (dilute, neutral / alkaline)
- Jones reagent

Quick Revision - Alcohols

-OH Group
-OH

makes alcohols polar and H-bonding.

Solubility

Lower alcohols (C₁-C₃) are soluble in water.

Boiling Point

Increases with molecular mass (due to H-bonding).

Acidity

Alcohols are very weak acids.

pK_a ≈ 16-18

Stability

3° > 2° > 1°
(Carbocation stability)

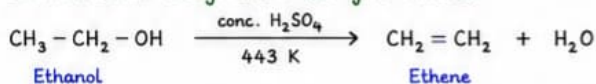


Understand → Practice →
Score High in NEET! ♥

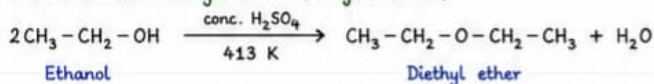
Dehydration of Alcohols (Elimination)

→ Removal of water from alcohols gives alkenes or ethers depending on conditions.

① Intramolecular Dehydration (to give Alkene)



② Intermolecular Dehydration (to give Ether)

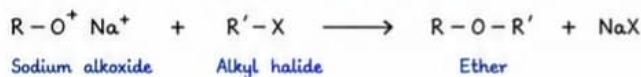


Reactivity Order of Different Compounds with HX

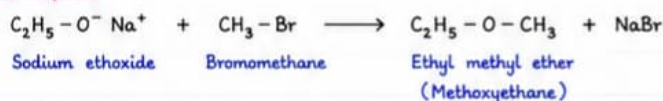


Preparation of Ethers :

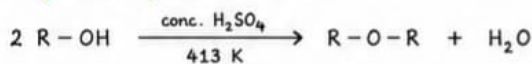
① Williamson Ether Synthesis (best method)



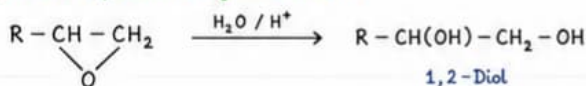
Example :



② Dehydration of Alcohols (413 K)



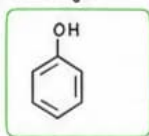
③ From Epoxides (cyclic ethers)



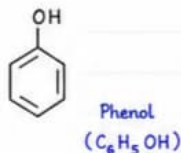
2. Phenols (Ar-OH group)

→ Phenols contain -OH group directly attached to aromatic ring.

General Formula



Example :



Chemical Properties of Phenols :

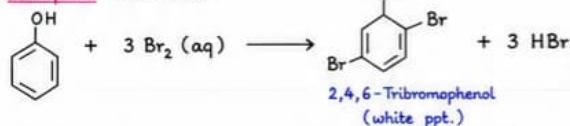
① Acidic Nature : (weak acid)



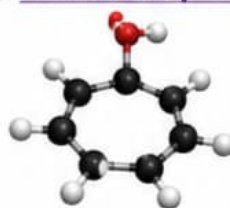
Stronger acid than alcohols (due to resonance stabilization of phenoxide ion).

② Electrophilic Substitution (at o- & p-positions)

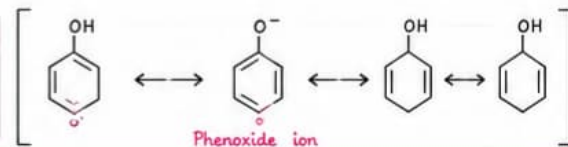
Example : Bromination



3D Structure of Phenol



-OH group is planar with the benzene ring due to resonance.



③ Kolbe's Reaction (Carboxylation)



Alcohols have -OH group on aliphatic carbon.



Dehydration gives alkenes (443 K) or ethers (413 K).



Ethers are less polar than alcohols.



Phenols are more acidic than alcohols due to resonance.



Phenols undergo electrophilic substitution at o- & p-positions.



Chemical Properties of Phenols (Contd.):

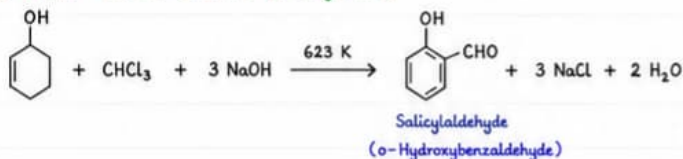
④ Reaction with Neutral FeCl_3 (Test for Phenol)



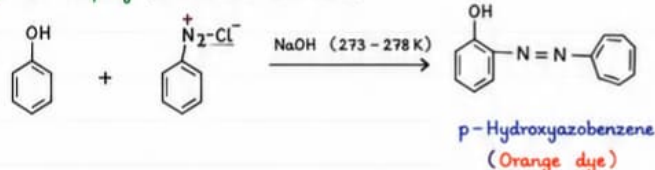
⑤ Nitration:



⑥ Reimer-Tiemann Reaction (Formylation)



⑦ Azo Coupling (with Diazonium Salts)



Phenols are more reactive than alcohols towards E.S. reactions due to resonance stabilization of phenoxide ion.

Comparison of Alcohols, Phenols and Ethers:

Property	Alcohols	Phenols	Ethers
Functional Group	-OH attached to sp^3 carbon	-OH attached to aromatic (sp^2) carbon	$\text{R}-\text{O}-\text{R}'$
Polarity	Polar (H-bonding)	Polar (H-bonding)	Less polar (No H-bonding with water)
Boiling Point	Highest (due to H-bonding)	Higher than ethers (lower than alcohols)	Lowest
Solubility in Water	Lower members soluble	Lower members soluble	Lower members (up to 4C) soluble
Acidity	Very weak acids ($\text{pK}_a \sim 16-18$)	More acidic than alcohols ($\text{pK}_a \sim 10$)	Not acidic
Reactivity	Less reactive	More reactive (E.S. reactions easy)	Least reactive (inert towards most reagents)

How to Distinguish? (Simple Tests)

Compound	Test	Observation
Alcohol	Add Na metal	Brisk effervescence of H_2 gas.
Phenol	Add neutral FeCl_3 solution	Violet / Purple colour appears.
Ether	Add Na metal	No effervescence.

Important Reactions at a Glance:

- Alcohol + $\text{HX} \longrightarrow$ Alkyl halide + H_2O
- Alcohol $\xrightarrow{[O]}$ Aldehyde / Ketone $\xrightarrow{[O]}$ Carboxylic acid
- Phenol + $\text{Br}_2(\text{aq}) \longrightarrow$ 2,4,6-Tribromophenol (white ppt.)
- Williamson synthesis $\longrightarrow$ Best method for ethers
- Phenol + $\text{FeCl}_3 \longrightarrow$ Violet colour (Test for phenol)

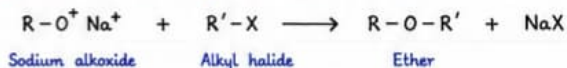
3. Ethers ($\text{R}-\text{O}-\text{R}'$)

General Formula: $\text{R}-\text{O}-\text{R}'$

(where R and R' may be same or different)

Preparation of Ethers:

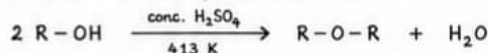
① Williamson Ether Synthesis (Best Method)



Example:



② From Alcohols (Dehydration, 413 K)



③ From Epoxides



Physical Properties of Ethers:

- ★ Lower members (up to 4 carbons) are soluble in water.
- ★ Higher members are less soluble in water.
- ★ They are volatile, inflammable and have characteristic pleasant smell.
- ★ Ethers do not form H-bonding with water (less polar than alcohols).



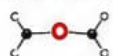
Uses:

- ★ Methanol: Used as a solvent and in the manufacture of formaldehyde.
- ★ Ethanol: Used in medicines, perfumes, as a solvent and as fuel (spirit).
- ★ Phenol: Used as antiseptic, in phenolic resins, plastics, dyes and drugs.
- ★ Ethers: Used as solvents (organic reactions), in anaesthesia (diethyl ether).



Quick Revision - One Line Points

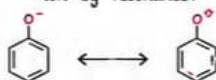
- ✓ Alcohols have -OH on sp^3 carbon, phenols on aromatic ring and ethers have $\text{R}-\text{O}-\text{R}$ linkage.



- ✓ Boiling point order: Ethers < Alcohols < Phenols (generally).



- ✓ Phenols are more acidic than alcohols due to stabilization of phenoxide ion by resonance.



- ✓ Ethers are prepared best by Williamson synthesis.



- ✓ Remember: Reactivity order with HX - $3^\circ > 2^\circ > 1^\circ$ alcohol. Phenols do electrophilic substitution easily.

